QR CODE **WATCH THE VIDEO BEFORE YOU SOLVE** YouTube • Lesson 6

RULE · Vertex form (completing the square)

$a(x-h)^2+k, \quad \text{vertex} = (h,k)$

$y = x^2 - 6x + 5 \implies y = (x-3)^2 - 4, \ \text{vertex } (3,-4)$

RULE · Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

Derived from completing the square on $ax^2+bx+c=0$. Use when factoring isn't clean.

RULE · Applying the formula – write both roots

$x^2+4x+1=0$: $a=1, b=4, c=1 \implies x = \dfrac{-4 \pm \sqrt{12}}{2}$

$x = -2+\sqrt{3}$ or $x = -2-\sqrt{3}$

A quadratic has up to two roots – write both unless the question asks for only one.

RULES SHEET – 2/2

RULE · Watch the sign on $-b$

If b is negative, $-b$ becomes positive – substitute carefully.

$x^2 - 6x + 4 = 0$: $b = -6 \implies -b = 6$

$$x = \frac{6 \pm \sqrt{36 - 16}}{2} = \frac{6 \pm \sqrt{20}}{2}$$

$x = 3 + \sqrt{5}$ or $x = 3 - \sqrt{5}$